Sahaj Bhadja

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Summary

Software Engineer with 2+ years building scalable **Java** and **Python** services, production LLM workflows, and SQL-backed systems, with object-oriented design, automated test suites, CI/CD, and **monitoring** for reliable cloud platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured fault-tolerant asynchronous **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Designed a multi-tenant platform on Azure SQL with row-level security for 500+ users, tuning queries to cut dashboard latency from 1.2s to 350ms, and shipped pipelines at 99%+ **reliability**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a production **AI-powered** platform with React and Flask on AWS that routes requests across GPT-4o, Claude, and Qwen-VL, serving 150+ instructors and 10K+ daily requests at peak load.
- Logged **per-call token usage and cost** for every model invocation and used the metrics to tune workflows, reaching 90%+ grading consistency while cutting instructor feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed object-oriented **Java** and Python modules with Singleton and Factory patterns for cross-platform microservices, and wrote 20+ **automated test suites** in Mockito and PyTest that ran nightly with coverage reports.
- Optimized large-scale API integrations with pagination, field filtering, and caching, improving response times 30%, and redesigned AWS Glue ETL boosting pipeline efficiency 68% in Agile sprints with Microsoft stakeholders.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **Java Spring Boot** microservices and a React frontend for a CRM platform covering leads, customer records, billing, and notifications, syncing 2M+ records across locations with real-time WebSocket updates.
- Tracked errors and performance with Sentry, Azure Monitor, and Kibana **telemetry**, gated releases on automated tests in Jenkins CI/CD, and used analytics to lift lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built booking, pricing, and coupon **REST APIs** with Spring Boot and Flask plus a React TypeScript dashboard, keeping responses under 200ms across 50K+ records with Firebase real-time updates.
- Integrated multi-currency card and wallet payments through idempotent webhooks, automating refunds and disputes to cut manual handling about 70%, and presented the end-to-end architecture to the client before implementation.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, HTML
Backend & AI: Spring Boot, FastAPI, Flask, REST APIs, Microservices, LLM Systems, Prompt Engineering, RAG
Frontend & Data: React, Next.js, PostgreSQL, Azure SQL, MySQL, ETL Pipelines, Data Structures
Quality & Observability: Mockito, PyTest, Jest, Selenium, Unit Testing, Code Coverage, Sentry, Kibana, Azure Monitor
Cloud & Practices: AWS, Azure, Docker, Kubernetes, CI/CD, Jenkins, Git, Object-Oriented Design, Agile

Projects

CRM Case Triage Agent | *Java, Spring Boot, LLM Tool Calling, PostgreSQL, JUnit*

- Designed an **AI agent** that reads incoming support cases, looks up account history through typed tool calls, suggests priority and routing with cited reasons, and requires human approval, with an **evaluation suite**.

Service Telemetry Dashboard | *Python, OpenTelemetry, Prometheus, Grafana, Docker*

- Designed **telemetry** for a microservice demo with traces, latency and error-rate metrics, and alert rules in Grafana, plus automated load tests that check **service-level objectives** before each release.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, REST APIs*

- Built a **retrieval** platform with LLaMA-2, transformer embeddings, and FAISS over 50K+ documents, using incremental indexing to cut retrieval latency 40% while sustaining 95% query accuracy for 200+ users.